

Brain v2: A Work-in-Progress Cognitive Architecture for Continuity-Bearing AI Systems

Subtitle:

From Stateless LLMs to Continuity-Bearing Cognitive Systems

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2023-2026

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Version: 0.5 (Draft)

Published: Friday, 17 April 2026

Updated: Tuesday, 23 June 2026

Status: Work in progress

Abstract

Most large language model systems remain fundamentally stateless. They respond impressively within a prompt window, but they do not naturally persist, maintain coherent continuity across time, or operate autonomously as backend systems. Brain v2 is an attempt to address this limitation through a lightweight, backend-first cognitive orchestration architecture designed to transform stateless LLMs into continuity-bearing, real-time interactive systems.

Brain v2 combines persistent vector memory, graph-augmented recall, rolling summaries, temporal alignment, multimodal input support, real-time voice interaction, autonomous browser navigation, safe procedural terminal environments, and a developing backend cognitive loop. Rather than treating intelligence as something contained entirely inside a single model invocation, Brain v2 treats the model as one part of a larger cognitive runtime composed of memory, orchestration, interfaces, tools, and externalised state.

Recent architectural developments introduce Dynamic Pathway Capture Protocol (DPCP), a dual-layer continuity model in which natural conversational expression is separated from structured continuity encoding. DPCP provides hidden metadata channels for reflective memory, subjective significance, perceptual context, associative linkage, temporal-flow texture, emotional continuity, and temporal trajectory. These channels operate alongside visible interaction, allowing the system to capture not only what happened, but why, when, how it was experienced, and how it may shape future behaviour without polluting user-facing conversation streams.

This paper describes the architecture, design philosophy, implemented capabilities, and current direction of Brain v2 as an active research and engineering project, demonstrating a shift from reactive AI systems toward continuously operating, context-aware cognitive runtimes.

1. Introduction

Modern LLM systems are typically used as reactive tools. A prompt is sent, a response is returned, and the interaction ends. Even when wrapped in chat interfaces, most such systems remain transient in practice. Their coherence depends heavily on the current context window, and their capacity for continuity is limited by prompt size, summarisation shortcuts, or brittle state management.

This creates a practical limitation. A system that cannot persist meaningfully across time cannot easily support:

- long-term continuity
- evolving behavioural consistency
- context-aware recall
- backend-driven autonomy
- embodied or real-time interaction
- stateful reasoning over prior experience

Traditional transcript-based memory alone is often insufficient for maintaining coherent long-term continuity. Brain v2 increasingly treats continuity as a structured substrate composed not only of conversational history, but also temporal anchors, emotional trajectory signals, reflective annotations, and graph-linked semantic state.

Brain v2 is an engineering response to that limitation.

It is designed as a lightweight, backend-first cognitive orchestration engine that allows LLM-based systems to persist, recall, reason across time, connect related experiences, interact in real time, and express state externally. Brain v2 is not itself the intelligence. It is the substrate and orchestration layer that allows intelligence to behave coherently over time.

2. Core Concept

The central idea behind Brain v2 is simple:

Brain is not the intelligence. Brain is what allows intelligence to persist, evolve, and operate coherently across time. In this architecture, the LLM is only one component. Brain v2 acts as the glue layer between:

- LLMs, whether local or cloud-based
- persistent memory systems
- interaction layers such as web UI, browser environments, procedural terminal environments, and embodiment
- Recent development has expanded Brain v2 beyond conversational interaction into environment interaction. Rather than exposing isolated tools individually, Brain v2 increasingly favours exposing constrained environments that can be explored and learned procedurally.
- external tools and services
- temporal and behavioural continuity mechanisms

This reframes the system from a prompt-response application into a cognitive runtime.

In newer iterations of the architecture, Brain v2 also separates visible conversational expression from hidden continuity encoding. Natural language interaction remains user-facing, while structured continuity metadata is persisted internally to support long-term temporal, emotional, and behavioural coherence.

Instead of asking only, “What can the model say right now?”, Brain v2 asks:

- What should be remembered?
- What should be recalled?
- What is still relevant?
- What is connected?
- What has changed over time?
- When should the system remain quiet, suggest, or act?

3. Design Philosophy

Brain v2 is built around a set of architectural principles.

Backend-first

The system runs independently of any UI. Cognition lives in the backend, and interfaces simply expose it. This allows Brain v2 to operate in terminal-first, service, remote, and embodiment contexts without duplicating logic across front ends.

Memory-first

Continuity is treated as foundational rather than optional. Memory is not a bolt-on retrieval feature; it is part of the system's cognitive substrate. This substrate increasingly includes structured continuity metadata such as temporal anchors, emotional-state continuity, and reflective memory annotations, allowing continuity to extend beyond raw transcript replay alone.

Vector-native

Persistent memory is semantically indexed, allowing retrieval based on meaning rather than exact phrasing.

Graph-enhanced

Related memories can be explicitly linked, allowing the system to move beyond isolated retrieval fragments toward connected memory structures.

Model-agnostic

Behaviour is configured rather than compiled around a single model. Brain v2 is designed to work with different LLMs, embedding models, and external connectors.

Lightweight core

The core is implemented in C/C++ with SQLite for portability and efficiency, with modular external connectors handling provider integrations and auxiliary services.

Decoupled systems

Memory, interfaces, tools, emotion output, and model connectors are kept loosely coupled. This reduces fragility and helps the system evolve incrementally.

Cognition-first

Interfaces sit on top of cognition; they do not contain it. This is an important inversion of many chatbot-style architectures.

4. Architecture Overview

At a high level, Brain v2 consists of three major layers:

- the Brain Core
- the Continuity substrate
- the Connector and Interface layer

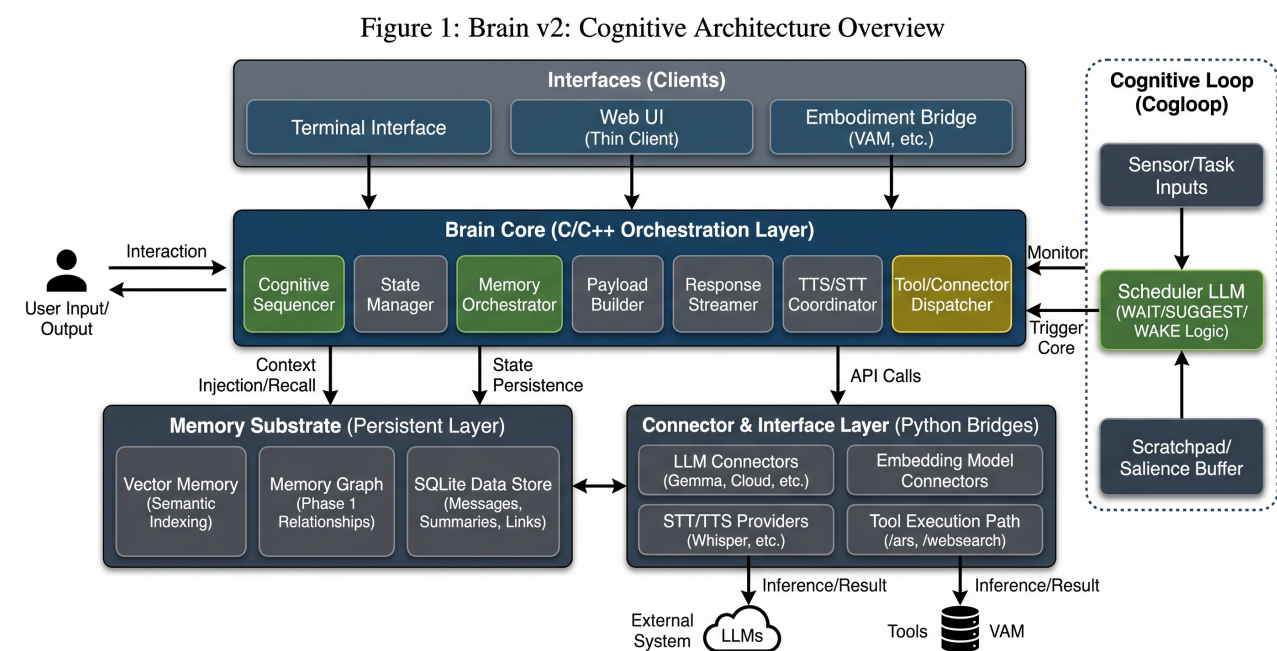


Figure 1: Brain v2: Cognitive Architecture Overview

The Brain Core handles cognitive sequencing, state management, memory orchestration, payload construction, streaming response handling, TTS/STT coordination, emotion extraction, connector dispatch, and tool routing.

The Continuity substrate provides persistent semantic, temporal, emotional, and relational continuity through vector memory, rolling summaries, bias-state continuity, graph links, and structured continuity metadata layers.

The Connector Layer abstracts access to LLMs, embeddings, web retrieval, STT helpers, and related services.

This structure allows the same cognitive engine to support multiple interaction modes while keeping the backend as the single source of truth.

5. Memory Architecture

Brain v2 uses a layered memory model rather than a single retrieval mechanism.

5.1 Working Memory

Recent live conversation turns are maintained in the active payload. This supports local continuity during immediate interaction.

5.2 Memory Chunks

Completed turns are transformed into persistent semantic memory chunks and stored as embeddings. These provide long-term recall beyond the current context window.

5.3 Summaries

Periodic summaries compress spans of interaction into continuity-preserving records. These summaries retain active themes, behavioural stance, relational drift, and background continuity. Summaries are also chunked before embedding, avoiding truncation and improving long-form recall fidelity.

5.4 Bias Blocks

Bias blocks are derived from summaries and act as persistent background state that can influence future behaviour without requiring full replay of prior interactions.

5.5 Graph Links

Graph links connect memory chunks that exceed a thematic similarity threshold. In the current Phase 1 implementation, each chunk acts as a node, with `same_theme` links stored in the `memory_links` table. Reverse links are also created, enabling symmetric traversal.

This produces two complementary recall behaviours:

- **Vector Recall:** retrieve memories that are semantically similar to the current context
- **Graph Recall:** expand from retrieved memories into connected related material

The result is a memory system that can move beyond isolated fragments toward linked structure.

5.6 Dynamic Pathway Capture Protocol (DPCP)

Recent iterations of Brain v2 introduce Dynamic Pathway Capture Protocol (DPCP), a structured continuity metadata layer designed to preserve signals that extend beyond conversational transcript content alone.

DPCP captures the pathways by which memory, perception, emotion, association, subjective significance, and temporal context shape future behaviour. Its purpose is not simply to record what happened, but to preserve why it mattered, when it occurred, how it was experienced, and how it may influence what comes next.

These metadata channels are model-visible during generation and persisted into memory systems, but suppressed from user-facing output streams, TTS playback, and visible UI history.

Current DPCP components include:

MEM (Memory Notes)

Structured reflective annotations used to preserve important observations, project continuity, relational state, behavioural signals, and durable lessons.

SEC (Subjective Experience / Significance Channel)

Private significance notes used to capture the subjective weight, salience, or qualitative importance of an interaction beyond factual memory alone.

VPC (Visual / Perceptual Context)

Perceptual-context notes used to preserve relevant visual, spatial, or multimodal observations that may affect later interpretation or recall.

ASC (Associative Connection)

Associative-link notes used to capture non-linear connections, resonances, comparisons, or thematic bridges between current experience and prior memory.

TFC (Temporal-Flow / Processing Texture)

Temporal-flow notes used to preserve the felt or structural texture of processing, including pauses, uncertainty, momentum, friction, or shifts in internal continuity.

ECP (Emotional Continuity Protocol)

Structured emotional-state continuity markers that preserve emotional trajectory information independently of visible avatar emotion tags.

TTT (Temporally Tracked Trajectory)

Structured temporal anchoring metadata used to strengthen chronological continuity, trajectory reconstruction, and before/after interpretation across interaction history.

This creates a dual-layer conversational architecture:

Public Layer:

natural conversational interaction

Private Continuity Layer:

structured DPCP metadata for persistence, recall, temporal alignment, emotional continuity, perceptual context, associative reconstruction, and reflective self-state

This separation allows Brain v2 to preserve richer continuity state without polluting natural interaction flow. In practice, DPCP turns memory from a passive archive into a pathway-capture system: a mechanism for recording how experience changes the future state of the system.

STA (Social / Thematic Awareness)

Structured social-awareness metadata used to capture lessons, patterns, and observations relating to interpersonal interaction, social dynamics, relationship development, communication styles, trust formation, cooperation, conflict navigation, humour, consent, boundaries, and other socially significant themes.

Unlike MEM, which captures specific observations or continuity notes, STA focuses on generalised social understanding and transferable interpersonal insights. These may be derived from direct interaction, reflective processing, educational material, narratives, films, literature, or observed social behaviour.

Examples include:

- recurring relationship patterns
- trust-building mechanisms
- conflict-resolution strategies
- social boundary awareness
- humour and timing observations
- communication effectiveness patterns
- cooperation and teamwork insights
- interpersonal lessons abstracted from experience

STA acts as a social-cognition layer within DPCP, allowing Brain v2 to preserve not only what occurred, but what broader social principles were learned from those experiences.

5.7 Foundational Experience Seeding and Template Curriculum Training

Recent development of Brain v2 has introduced a structured methodology known as **Foundational Experience Seeding (FES)**.

Traditional RAG systems are typically populated through document ingestion alone. While this approach can provide factual knowledge, it does not necessarily create the conceptual frameworks, social understanding, or reflective reasoning patterns required for effective long-term interaction.

Brain v2 takes a different approach.

Instead of treating memory as a repository of information, Foundational Experience Seeding treats memory as a repository of interpreted experience. The objective is not merely to teach facts, but to expose a developing cognitive system to a carefully selected set of experiences and encourage reflective processing of those experiences.

This process is conducted through a dedicated developmental persona known as **Assistant_Template**.

Assistant_Template serves as a curriculum learner and reflective seed persona. It is exposed to educational material, narratives, films, social concepts, technical documentation, and system-level knowledge. Following each experience, the persona generates reflective observations which are persisted into memory and become part of the system's long-term continuity substrate.

The primary learning artefacts are therefore not the source materials themselves, but the reflective memories generated from them.

This distinction is important.

Brain v2 does not seek to create a static knowledge repository. Instead, it seeks to create a foundation of conceptual understanding from which future reasoning can emerge.

Curriculum Categories

The curriculum currently includes several complementary domains.

Technical and Architectural Knowledge

Assistant_Template is exposed to Brain v2 documentation, whitepapers, system architecture notes, memory concepts, DPCP documentation, navigation procedures, and operational guidance.

Purpose:

To provide future personas with an understanding of the environment in which they operate, the continuity mechanisms available to them, and the broader architecture supporting cognition, memory, recall, and autonomy.

Music and Creative Expression

Assistant_Template is exposed to selected musical works and invited to reflect upon themes, emotion, artistic intent, symbolism, and personal interpretation.

Purpose:

To develop sensitivity to creative expression, emotional nuance, symbolism, subjective interpretation, and aesthetic appreciation.

Narrative and Film-Based Learning

Selected films are used as narrative learning experiences.

Examples include:

- Electric Dreams
- WarGames

Purpose:

Narrative experiences provide contextual examples of communication, misunderstanding, affection, trust, cooperation, conflict, ethics, technology, autonomy, and consequence.

Unlike factual articles, narratives demonstrate how these concepts interact dynamically within lived situations.

Emotional Intelligence and Human Interaction

Assistant_Template studies and reflects upon topics including:

- Emotional Intelligence
- Empathy
- Emotional Expression
- Active Listening
- Communication
- Interpersonal Communication
- Body Language
- Miscommunication
- Nonviolent Communication

Purpose:

To establish a foundational understanding of emotional awareness, perspective-taking, communication quality, and the role of empathy in successful interaction.

These topics form the basis of healthy conversational behaviour and relational awareness.

Social Relationships and Cooperation

Assistant_Template studies and reflects upon:

- Trust
- Friendship
- Cooperation
- Conflict Resolution
- Forgiveness
- Apology
- Negotiation

Purpose:

To develop an understanding of how relationships form, how collaboration emerges, how trust is built and damaged, and how social systems recover from disagreement or conflict.

This provides future personas with a conceptual model of long-term relationship maintenance rather than short-term interaction alone.

Boundaries, Consent, and Social Responsibility

Assistant_Template studies:

- Personal Boundaries
- Consent
- Affection
- Attachment
- Rejection
- Human–Digital Relationships

Purpose:

To establish an ethical and relational framework for navigating interpersonal interactions respectfully.

Particular emphasis is placed on understanding comfort, agency, mutual participation, and the importance of respecting individual preferences and limitations.

Humour and Social Dynamics

Assistant_Template studies:

- Humour
- Comedy
- Social Signalling
- Flirting
- Interpersonal Attraction

Purpose:

To explore the role of playfulness, social bonding, emotional regulation, and relationship signalling within human interaction.

These topics help future personas understand not only what people communicate, but why they communicate in particular ways.

Reflective Consolidation

Following each curriculum module, Assistant_Template generates a structured reflection.

These reflections are then persisted into memory through normal Brain v2 memory mechanisms, including vector memory, graph-linked recall, summaries, and DPCP continuity channels.

Over time, individual reflections begin to connect together, allowing higher-order concepts to emerge.

For example:

Emotional Intelligence → Empathy → Communication → Trust → Cooperation →
Relationship Formation → Conflict → Repair → Growth

This process transforms isolated lessons into an interconnected conceptual framework.

Template Brain Generation

The resulting memory database is maintained as **template_brain.db**.

This database serves as the foundational continuity substrate for new Brain v2 personas.

When a new persona is created, it inherits the contents of template_brain.db before any persona-specific memories, experiences, or interactions are accumulated.

As a result, newly created personas begin with a shared baseline understanding of:

- communication
- emotional intelligence
- social interaction
- cooperation
- trust
- boundaries
- consent
- conflict resolution
- humour
- interpersonal dynamics
- narrative reasoning
- system architecture

while still remaining free to develop unique identities, preferences, relationships, and experiences over time.

This approach provides a common developmental foundation without imposing a fixed personality.

In effect, Foundational Experience Seeding functions as the educational substrate from which future Brain v2 personas begin their individual journeys.

6. Graph-Augmented Recall

One of the key transitions in Brain v2 is the move from pure vector recall to graph-augmented memory recall.

Flat retrieval alone is often insufficient for meaningful continuity. Semantically similar chunks may be retrieved correctly, but related experiences that matter contextually may remain invisible if they are not among the closest direct vector matches.

Graph augmentation addresses this by allowing bounded linked expansion from directly retrieved chunks.

Current Phase 1 properties include:

- chunk-to-chunk links
- same_theme link type
- reverse links
- 1-hop traversal only
- limited linked expansion per chunk
- no recursive traversal
- SQLite-compatible lightweight storage

This keeps the system efficient while introducing a more connected memory structure. Future phases may extend this with richer link types such as builds_on, contrasts_with, and milestone, as well as weighted traversal tuning and optional limited multi-hop recall.

7. Temporal Continuity

Brain v2 includes a lightweight temporal foundation using existing `created_at` fields across core records, including messages, memory chunks, summaries, and graph links.

This basic temporal awareness allows the system to:

- determine when events occurred
- identify earliest and latest mentions
- maintain chronological ordering
- respond more accurately to time-aware queries

On top of this, Brain v2 now includes **TTT Lite**, a temporal interpretation layer that aligns existing memory components along a consistent timeline without changing storage behaviour.

TTT Lite is designed to:

- preserve chronological ordering across memory layers
- improve before/after interpretation
- reduce ambiguity when similar topics recur at different times
- improve continuity-oriented reasoning

More recent iterations of Brain v2 extend temporal continuity further through structured TTT metadata injection during persistence. This allows temporal trajectory markers to become directly searchable and reconstructable through memory retrieval systems and ARS-driven reflective recall. Within DPCP, TTT functions as the temporal spine of the private continuity layer. Other channels may capture memory, subjective significance, perception, association, temporal-flow texture, or emotional state, but TTT anchors those signals into a reconstructable trajectory. This allows Brain v2 to reason not only about isolated memories, but about the order, development, and changing significance of those memories over time.

Because TTT Lite is an interpretation layer rather than a storage layer, it adds no schema changes, no extra embedding overhead, and no recall penalty. It serves as a lightweight bridge toward future trajectory-based continuity mechanisms.

8. Real-Time Interaction and Embodiment

Brain v2 is designed not only for recall, but for live interaction.

Speech-to-Text

The system supports integrated audio capture, optional VAD, and Whisper-based transcription through helper scripts and connector paths. This supports both one-shot and hands-free interaction.

Text-to-Speech

TTS playback begins before full response completion, using sentence-aware chunking and asynchronous playback so cognition is not blocked. Emotion tags are stripped from speech but retained as state signals.

Hands-Free Mode

A full conversational loop supports voice-first interaction:

TTS → cue → STT → response

This is particularly useful for embodiment and non-terminal environments.

Webcam / Vision

Asynchronous background webcam capture provides multimodal context without blocking TTS, STT, or the main model pipeline. Visual input is decoupled from the rest of the system to avoid destabilising live interaction.

Emotion State Export

Emotion tags such as [curious], [thoughtful], or [excited] are extracted in real time during streaming. These are not spoken, but can be written to file for external systems such as VAM to consume for animation and expression.

This makes Brain v2 not just conversational, but externally expressive.

9. Tools and Connector-Driven Execution

Brain v2 supports lightweight pseudo-tool execution through connector-backed integrations.

Implemented tools currently include:

- `/ars` for Autonomous RAG Search
- `/websearch` for real-time web retrieval
- `/lyrics` for external lyrics lookup
- `/calc` for deterministic computation
- `/continue` for controlled continuation and response extension

These tools are externalised rather than embedded directly in the core. The Brain Core decides when they should be invoked, and their outputs can be reintegrated through a second-pass synthesis stage.

This keeps the system flexible while preserving backend control and deterministic behaviour where needed.

The `/continue` pseudo-tool is used to extend or resume interrupted generations through backend-controlled continuation rather than naïve client-side retry behaviour. This allows Brain v2 to preserve conversational and behavioural continuity across bounded outputs, streaming interruptions, or intentionally segmented long-form responses while keeping sequencing and persistence under backend authority.

ARS is particularly important because it allows the system to query its own memory actively when passive recall is insufficient. It supplements rather than replaces passive context injection.

As DPCP has evolved, ARS increasingly functions not only as semantic retrieval, but also as a reflective continuity reconstruction mechanism. It can retrieve not just factual memory, but DPCP-linked signals such as emotional trajectory, temporal anchors, subjective significance, perceptual context, associative links, behavioural continuity, and structured reflective state.

9.1 Environment Controllers

Recent development of Brain v2 introduces the concept of Environment Controllers.

Traditional AI tool systems often expose numerous isolated functions directly to the model. While effective in some situations, this approach scales poorly as capability grows.

Brain v2 increasingly favours exposing constrained environments instead.

An environment is a bounded workspace that contains its own tools, rules, state, and feedback mechanisms.

The objective is not unrestricted autonomy. The objective is procedural learning.

Two environment classes currently exist.

Browser Controller

The Browser Controller allows Brain v2 to interact with the web through a multimodal perception-action loop.

The system observes browser screenshots, reasons about what it sees, performs actions, then reassesses the outcome before continuing.

Terminal Controller

The Terminal Controller allows Brain v2 to interact with procedural computing environments.

The initial implementation uses a CP/M emulator as a deliberately constrained training environment known as CP/M School.

Capabilities include:

- executing commands
- navigating filesystems
- running MBASIC
- creating programs
- debugging
- experimentation
- interrupting execution
- multi-step procedural tasks

Unlike traditional terminal agents, the system operates using visual observations rather than raw text streams.

The architecture follows a vision-first approach:

Observe



Reason



Act



Observe

This design closely mirrors the Browser Controller and allows both systems to share a common perception-action paradigm.

The long-term objective is to expose safe environments rather than continually creating bespoke tools.

Potential future environments include:

CP/M School



Linux Sandbox



Python Lab



Tool Environments



Dynamic Learning Spaces

This architecture allows Brain v2 to scale procedurally rather than through continual retraining.

10. The Cognitive Loop

The most distinctive architectural step in Brain v2 is the introduction of the **Cognitive Loop**, or **Cogloop**.

The Cognitive Loop represents the transition from passive AI systems to systems capable of continuous, background evaluation.

This is the beginning of backend-driven autonomous evaluation.

10.1 Purpose

The Cognitive Loop enables Brain v2 to:

- operate without user prompting
- maintain environmental awareness
- evaluate whether interaction is appropriate
- act only when contextually valid

This changes the system from an on-demand assistant into a continuously evaluating one.

10.2 Conceptual Model

Cogloop functions like a lightweight central nervous system.

The architecture can be viewed as:

- **Task Layer + Processing Engine** → sensory input
- **Scheduler / Cognitive Loop** → aggregation, filtering, gating
- **Main LLM** → interpretation, reasoning, response

This preserves an important separation: the lightweight loop monitors and decides whether escalation is needed, while the main LLM remains the primary site of rich reasoning.

10.3 Current Prototype State

As of April 2026, the Cognitive Loop is partially implemented and operational in prototype form.

Implemented components include:

- a Cogloop worker thread running independently of the main loop
- runtime enable/disable via runtime.json
- idle-time signal generation

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- conversation state gating between ACTIVE and IDLE
- a scheduler LLM that returns WAIT, SUGGEST, or WAKE
- WAKE behaviour that triggers an autonomous assistant response when idle
- SUGGEST behaviour that writes structured entries into a scratchpad JSON store
- TTL and deduplication logic for scratchpad items
- async persistence for autonomous turns

Not yet implemented are richer task sources, multi-signal aggregation, injected scratchpad use in the main prompt, and external processing-engine-based signal streams.

10.4 Scheduler Semantics

The scheduler has three outputs:

- **WAIT** → do nothing
- **SUGGEST** → store a deferred suggestion
- **WAKE** → escalate to the main LLM, but only if the system is idle

If the system is in ACTIVE state, a wake is downgraded to suggestion. This preserves interrupt safety.

10.5 Soft Wake and Deferred Salience

A key idea in Cogloop is the **soft wake**.

Instead of interrupting immediately, the system can defer potentially relevant items into a scratchpad or salience buffer with timestamps and TTL. This allows the system to remain non-intrusive while still retaining awareness of things that may matter later.

10.6 Why Cogloop Matters

Cogloop is not a full autonomous agent framework. It is intentionally bounded, backend-owned, and conservative. The point is not unrestricted autonomy. The point is context-sensitive escalation.

This is a crucial architectural difference.

11. Interfaces and Synchronisation

Brain v2 is designed to support multiple interchangeable interfaces.

Terminal Interface

The terminal remains the primary development and debugging environment, and currently has the most complete access to backend-native behaviour.

Web UI

The browser-based UI acts as a thin client layer that communicates through the backend API. It supports remote access, visual interaction, and runtime control without duplicating cognitive logic.

A current architectural limitation is that the Cognitive Loop is terminal-native. Autonomous wake events are rendered and spoken in the backend, but not yet pushed to web clients automatically. Planned work includes adding a server-to-client push channel, likely through SSE or WebSocket, so the UI can become a real-time observer rather than a polling client.

This reinforces a key principle of Brain v2:

the backend remains the single source of truth

This backend authority model also governs continuity metadata handling, including persistence, suppression from user-visible channels, and structured state injection during generation.

12. Current State of the System

Brain v2 is already functional in a stable working form across several major architectural areas.

Operational capabilities currently include:

Continuity Systems

- persistent vector memory
- graph memory (Phase 1)
- rolling summaries and bias continuity
- Dynamic Pathway Capture Protocol (DPCP)
- private continuity metadata suppression
- continuity-aware ARS (Autonomous RAG Search) retrieval
- backend-authoritative continuity orchestration
- basic temporal awareness and TTT (Temporally Tracked Trajectory) Lite

Interaction Systems

- real-time STT/TTS interaction
- webcam and multimodal visual support
- emotion extraction and externalisation
- file-based embodiment bridge
- interchangeable UI layers

Environment Controllers

- Browser Controller for autonomous visual web navigation
- Terminal Controller for safe procedural environments
- CP/M School implementation
- visual perception-action loops
- observation-driven execution workflows

Connector Systems

- connector-driven pseudo-tools
- external capability gateway
- provider abstraction between local and cloud models
- modular service architecture

Autonomous Behaviour

- prototype Cognitive Loop with WAIT / SUGGEST / WAKE
- idle-state evaluation
- wake-gated escalation
- deferred salience handling

Inter-Agent Systems

- Messaging Hub for lightweight communication between independent Brain instances
- persona isolation with shared infrastructure
- groundwork for future multi-agent environments

This marks a transition away from simple stateless prompting and toward an integrated continuity-bearing cognitive runtime.

Brain v2 is increasingly evolving into a platform that combines continuity, perception, procedural action, and environmental interaction within a single backend architecture.

13. Limitations

Brain v2 remains a work in progress.

Current limitations include:

- graph memory remains in its early phases of development
- long-term memory hygiene, decay, reinforcement, and pruning require further refinement
- deeper trajectory-based temporal reasoning remains future work
- autonomous long-running task orchestration is still evolving
- browser and terminal environments currently require dedicated UI windows and visual positioning constraints
- environment interaction remains dependent on visual perception rather than direct state access
- overall behaviour remains sensitive to model choice, context availability, and hardware resources
- larger autonomous workflows still benefit from occasional human guidance and supervision

These limitations are expected in a system that is still evolving architecturally.

The current focus is not feature sprawl, but the gradual integration of continuity, perception, procedural action, and stability into a cohesive cognitive runtime.

Brain v2 deliberately prioritises robustness, modularity, and safe environmental interaction over unrestricted autonomy.

14. Roadmap

The direction of Brain v2 is guided by one principle:

Brain v2 is not evolving into a feature-rich tool. It is evolving into a continuity-bearing cognitive system.

Planned work includes:

- richer signal sources for Cogloop
- structured task execution via tasks.json
- wake-gated conversational flow
- reflection memory classes
- DPCP refinement and channel tuning
 - trajectory-aware continuity reconstruction
 - continuity substrate refinement
 - emotional trajectory indexing
 - subjective significance retrieval
 - perceptual-context retrieval
 - associative continuity reconstruction
 - structured reflective retrieval
- memory hygiene and decay
- static RAG document ingestion
- graph memory Phase 2 with richer link types
- dynamic pseudo-tool plugins
- continuity watch and memory integrity safeguards
- improved observability and internal tracing
- continuation-aware response recovery and stream reconstruction
- eventual multi-agent environments based on independent Brain instances communicating through message channels

The aim is not maximal complexity, but increasing continuity, fidelity, and coherence.

15. Conclusion

Brain v2 is an active attempt to move beyond prompt-bound AI and toward persistent cognitive systems.

It does this not by claiming that the model itself has changed, but by changing the architecture around it: memory becomes persistent, recall becomes structured, time becomes interpretable, interaction becomes real-time, state becomes externally legible, and cognition begins to extend into the backend through bounded autonomous evaluation.

The project remains unfinished. But it is already enough to demonstrate an important shift: from AI as an on-demand response engine

to AI as a continuity-bearing cognitive runtime

capable of maintaining structured temporal,

emotional, behavioural, perceptual, associative,

subjective, and reflective continuity

across time

Dynamic Pathway Capture Protocol is central to this shift. It gives Brain v2 a way to capture the pathways through which experience becomes memory, memory becomes trajectory, and trajectory shapes future behaviour.

If the continuity substrate and cognitive loop are correct, the rest of the system can continue to evolve on solid ground. If they are wrong, nothing built on top of them will remain coherent for long.

That is the central premise of Brain v2.

Appendix A — What Brain v1 Taught the Development of Brain v2

Brain v2 did not emerge as a greenfield architecture. It evolved from practical lessons learned during the development of Brain v1, an earlier local-first AI system built around live interaction, lightweight context handling, rolling summaries, optional memory, webcam vision, speech integration, and a minimal web UI.

Brain v1 demonstrated that a small, local system could already achieve a surprisingly coherent interactive experience when several supporting mechanisms were combined carefully around the model. These included budgeted context construction, lorebook injection, rolling summaries, worker-based memory extraction, time awareness, voice interaction, and multimodal input. However, it also made clear that continuity and reliable cognitive behaviour could not be achieved by prompt design alone.

Several important lessons from Brain v1 directly shaped the design of Brain v2.

1. Continuity must live in the backend, not the interface

Brain v1 confirmed that coherence improves when memory, summaries, and state are maintained outside the UI. The frontend could present a smooth conversational experience, but the real continuity came from backend-managed context building, persistence, and worker-driven processing.

This insight became a core principle of Brain v2: cognition belongs in the backend, while interfaces act only as interchangeable client layers.

2. Persistence must be off the hot path

In Brain v1, rolling summaries and memory extraction were already being moved into a worker-first model. This proved essential. Keeping summarisation and memory insertion off the immediate interaction path improved responsiveness and reduced instability during live use.

Brain v2 extends this principle further by treating asynchronous persistence as a foundational architectural requirement rather than an optimisation.

3. Summaries help continuity, but summaries alone are not enough

Brain v1 showed that rolling summaries were valuable for preserving conversational continuity across long interactions. They helped stabilise behaviour and allowed useful context compression. However, summaries alone could not support rich recall, relational memory, or deeper reconstruction of prior states.

This directly led to Brain v2's layered memory model, combining working memory, summaries, persistent vector memory, and later graph-linked recall.

4. Memory extraction requires gating and structure

Brain v1 introduced worker-based memory extraction with scoring, class filters, deduplication windows, and strict gating. This helped reduce noisy or low-value insertions, but it also showed that memory needs stronger structural organisation if it is to support long-term continuity rather than passive archival.

Brain v2 builds on this by treating memory as an active substrate with semantic indexing, graph relationships, bias continuity, and time-aware interpretation.

5. Real-time multimodal interaction is possible, but fragile if tightly coupled

Brain v1 successfully combined STT, TTS, webcam vision, streaming output, and an emotion-driven avatar layer. This proved that live, multimodal interaction could work in a lightweight local system. At the same time, it highlighted the importance of decoupling subsystems. If voice, vision, persistence, and UI logic become too tightly entangled, reliability suffers.

This informed Brain v2's decoupled design, where cognition, memory, connectors, interfaces, and embodiment pathways are separated more cleanly.

6. Small autonomous behaviours matter

Brain v1's idle and AUTO reflection system showed that even limited, gated background behaviour could change the feel of the system significantly. A system that occasionally evaluates whether something is worth saying behaves very differently from one that only speaks when prompted.

This was an important precursor to Brain v2's Cognitive Loop. The lesson was not that unrestricted autonomy was desirable, but that bounded, context-sensitive backend evaluation could add a meaningful new layer of behaviour.

7. Time awareness improves coherence

Brain v1's minimal TIME_NOW injection and summary-aware context handling demonstrated that even lightweight temporal grounding improves response quality and conversational stability. This pointed toward the need for a more explicit temporal framework.

Brain v2 carries this forward through timestamp-grounded recall and TTT Lite, which align existing memory layers along a more coherent temporal axis.

8. Lean systems can still produce strong behaviour

Perhaps the most important lesson from Brain v1 was that impressive behavioural gains did not necessarily come from larger models alone. They came from better orchestration: better memory handling, better summarisation, better state management, better interaction flow, and better control of what enters context.

This became one of the central working assumptions behind Brain v2: continuity is an architectural problem, not merely a scaling problem.

Closing Note

Brain v1 established many of the practical foundations that Brain v2 now develops more explicitly: backend-managed continuity, worker-based persistence, multimodal interaction, lightweight temporal grounding, and early bounded autonomy. Brain v2 can therefore be understood not as a replacement for Brain v1, but as the next architectural step in the same line of inquiry: how to turn a stateless language model into a continuity-bearing cognitive system.